Department of Biosciences and Bioengineering
BT 416
Bioenvironmental Engineering

Date: 06/03/2026

Mid Semester Examination
Answer ALL questions

Total Marks: 30

1. Explain with a diagram breakpoint chlorination and its significance in water disinfection (3 marks)

2. Calculate the BOD_5 (in mg/L) of a sample at 20 °C temperature. The initial DO values are at saturation, and the sample dilution is 1:30 with seeded dilution water. The final DO of the seeded dilution water is 8 mg/L, and the final DO of the sample and seeded dilution water is 2 mg/L. The volume of the BOD test bottle is 300 mL. (3 marks)

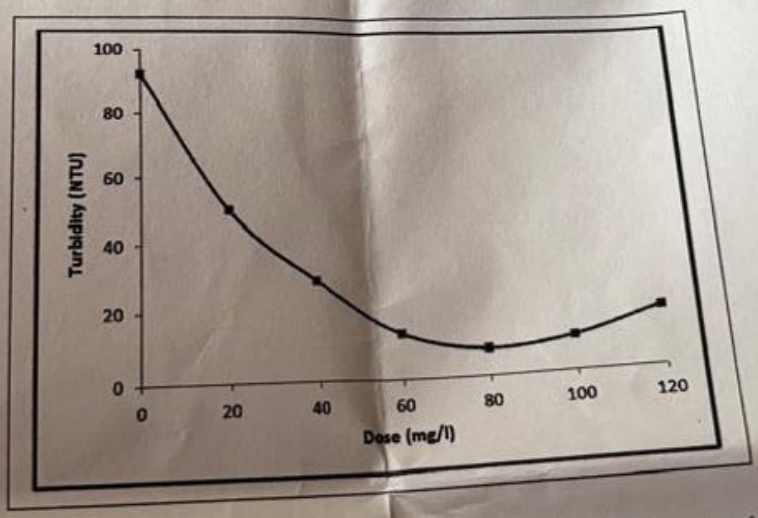
3. A water sample contains solid particles of spherical shape with 0.1 mm diameter and 1.05 g/cm^3 density. If the viscosity of water is 1.0 cP, determine the settling velocity (in cm/s) of the particles. (1 cP = 0.01 g/cm·s). (3 marks)

4. The following table presents the characteristics of wastewater collected from an industry:

pH	Suspended solids (mg/L)	BOD (mg/L)	COD (mg/L)	BOD load per kg of product	Other characteristics
4.5	6250	32000	47000	560	High TDS, Cl, SO_4

Draw a schematic showing the various unit operations for treating the industrial wastewater. (3 marks)

5. In an experiment to reduce the turbidity of a water sample, different doses of the coagulant aluminium sulfate were tested, and the results were obtained as shown in the following figure.



(a) How is the turbidity of water measured? What does the turbidity unit NTU stand for in the figure?
(b) Why does the turbidity decrease with an increase in the coagulant dose from 0 to 80 mg/L and gradually increase above 100 mg/L dose. Explain the underlying mechanism involved in each case. (4 marks)

6. Differentiate between Dot Blot Hybridization and Fluorescent In Situ Hybridization techniques for th enumeration of microorganisms in water. Show a schematic of the steps involved in each case. (4 marks

7. The following equation represents the relationship between the time (t) required for a 99.9 percent kill for chlorine dosage (C) of 0.05 mg/L at a temperature (T) of 5 °C and pH of 8.5.

$$\ln \frac{N_t}{N_0} = -10.48\, C^{1.20}\, t$$

Estimate the time required (in minutes) at 20 °C temperature. Given the values of the activation energy and gas constant are 26,800 J/mol and 8.3144 J/mol·K, respectively. (5 marks)

8. A waste stream with a dissolved oxygen concentration of 1.5 mg/L, a flow of 0.5 m^3/s, a temperature of 26 °C, and an ultimate biochemical oxygen demand (BOD_U) of 48 mg/L is discharged into a large stream with a reoxygenation constant value of 0.4 day^{-1} and a velocity of 0.85 m/s. The stream water flow is 2.2 m^3/s, with a saturated dissolved oxygen concentration of 12.0 mg/L, a temperature of 12 °C, and a BOD_U value of 13.6 mg/L. Calculate the dissolved oxygen concentration (in mg/L) at a distance of 48.3 km downstream from the waste organic discharge. Assume that the value of the deoxygenation constant is 0.2 day^{-1} and the initial oxygen deficit D_0 is 1.1 mg/L. (5 marks)